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Datasci 224 final project

Gaussian Process Regression applied to ecological model

Background

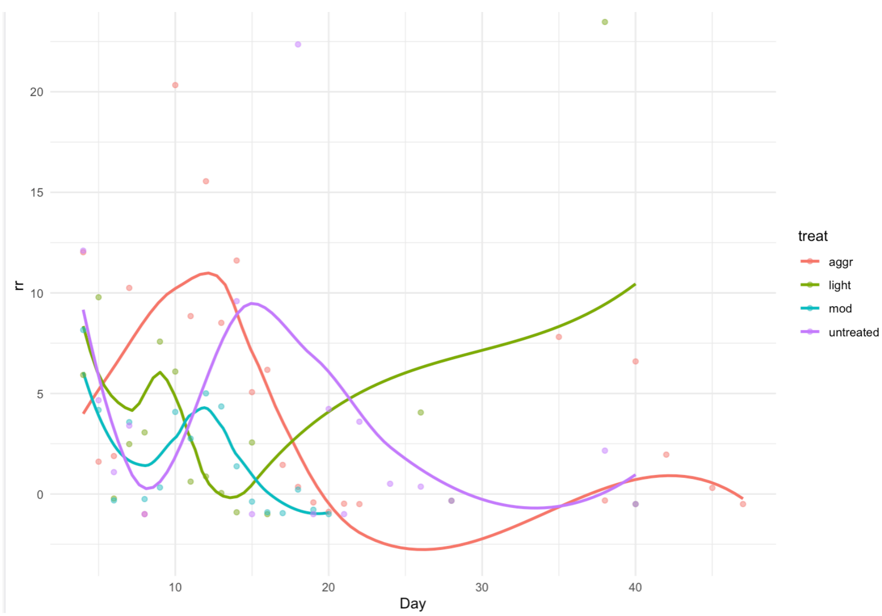
The dataset that I'm using is from an in vivo study, observing selection and relapse dynamics through longitudinal measurements in Plasmodium Chabaudi infected mice, artificially inoculated with susceptible and resistant genotypes, and intervened by varying treatment regimens by aggressiveness. (1) Plasmodium, being a eukaryotic pathogen behaves differently from bacterial pathogens in many ways, including mutagenesis, genotype interactions, drug evasion, etc. and they can have more sophisticated capabilities, like variability through expression. Nonetheless, it is a very elegant choice for selection analysis between genotypes due to several factors. First, the blood is a strictly sterile environment which means that there are (normally) no other pathogens in the environment interacting with the plasmodium or responding to antimicrobials thus shifting the ecosystem. In addition, it eliminates variability from spatial orientations of mutants like in clonal infections. Finally, Plasmodium is highly dependent on red blood cells (RBC) for survival and replication, which can be considered a singular resource and it significantly simplifies our models when considering resource mediated interactions. One of the more fascinating features of Plasmodium is that their cycle follows a circadian rhythm, producing discrete progression which is also convenient as data collection is also discrete. (3)

In the bloodstream, Plasmodium chabaudi undergoes its erythrocytic cycle, which is responsible for the clinical manifestations of malaria. Merozoites released from the liver invade RBCs and develop through the ring, trophozoite, and schizont stages. Each schizont produces multiple new merozoites inside the host RBC, which are released upon rupture of the RBC, allowing the infection to propagate. Within minutes after rupture, the merozoites either enter a new RBC (one RBC per merozoite) or die. This cycle typically repeats every 24 hours for P. chabaudi with synchronized bursting RBCs at nights, generating waves of parasitemia. (3)

How we define treatment aggressiveness is based on the mechanism of the antimicrobial, which can be categorized as concentration dependent mechanisms and time dependent mechanisms. Concentration dependent mechanism is when the intensity of antimicrobial effect is a continuous function of the drug's concentration. In this case aggressiveness is defined by mean plasma concentration of the drug maintained during treatment. In time dependent mechanisms the antimicrobial effect is present when the drug concentration is above a certain threshold and not present if below the threshold. This time, aggressiveness is defined by the duration that the concentration is kept above the threshold. In the case of this study, the antimalarial Pyrimethamine operates through a time dependent mechanism. This means at every observation we have a binary condition regarding treatment.

My aim was to attempt an equilibrium assessment of the system to find the territory where selection of the resistant strain becomes more likely and where our optimal strategy potentially lies. To achieve this we need to first model dS/dt and dR/dt and in terms of the relevant variables. Since data measurements are discrete and progression is biologically also partially discrete, it is more mathematically accurate to use difference equations instead of differential equations. I used an established ecological competition model as reference. In this analysis I model only the resistant strain in the untreated portion using gaussian process regression. I will be assessing the ecological representation of the difference equations.

Temporal trends of per capita change for the four treatment groups in the antibiotic resistant group



Algorithm

Gaussian Process Regression is a non-parametric Bayesian approach to regression that infers a posterior distribution over an infinite-dimensional space of functions. A Gaussian Process is a collection of random variables which in this case belong in the continuous space. Any finite subset of the space has a joint multivariate Gaussian distribution. Let $x \in \mathbb{R}^d$ denote a d-dimensional input vector. A continuous latent function f(x) is distributed as a Gaussian Process represented by

$$f(x) \sim GP(m(x), k(x, x'))$$

where the prior mean function is typically centered to zero (m(x)=0). Assuming an additive, independent and identically distributed (i.i.d.) Gaussian noise structure, the observation model is:$y = f(x) + \epsilon, \epsilon \sim N(0,\sigma_n^2)$

The covariance function $k(x, x')$ defines the similarity and smoothness between points. A standard choice is the Squared-Exponential (SE) or equivalently the Radial Basis Function (RBF) kernel:

$k(x_p, x_q) = \sigma_f^2 exp(-\frac{1}{2l^2} \parallel xp - xq \parallel^2)$ where $\theta = \sigma_f^2, l, \sigma_n^2$ represents the hyperparameters: signal variance σ_f^2 (vertical scale) and characteristic length-scale l (horizontal scale).

By computing a static weight vector $w = [K(X, X) + \sigma_n^2 I]^{-1} y$, the predictions simplify to a linear combination of the kernel evaluations: $f_* = K(X_*, X)w$. The diagonal entries of cov(f$_*$) provide the point-wise predictive variances.

A primary motivation for using GPR in this context is the ability to evaluate a biological system without being constrained by fixed assumptions about its functional form. A perceived biological mechanism might suggest a specific differential or parametric equation. If the underlying biological perception is slightly flawed or incomplete, a parametric model will exhibit significant bias.

GPR, being a non-parametric Bayesian method, avoids this limitation. Instead of estimating a fixed set of parameters for a single line or curve, it infers a full probability distribution over an infinite space of possible functions. This allows the model to purely adapt to the data, letting the empirical observations dictate the continuous progression across the domain.

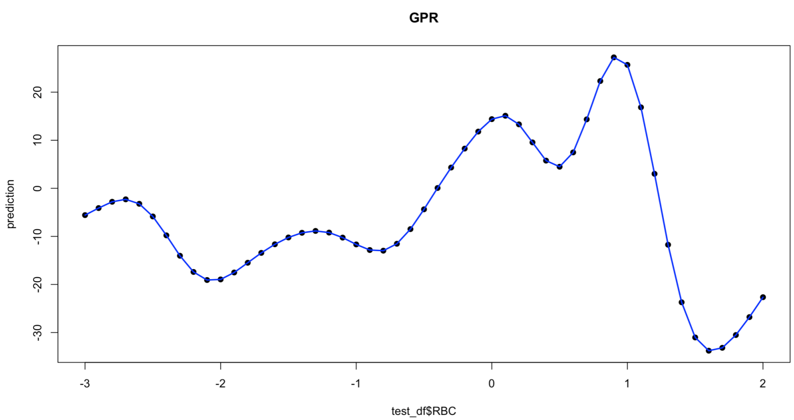
Results and Discussion

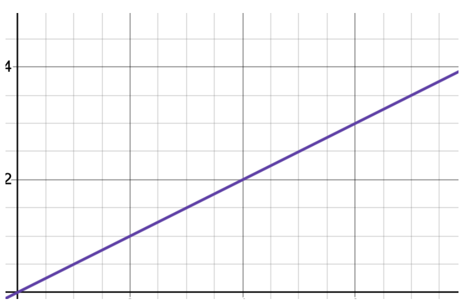
The predictive performance of the data-driven GPR model was compared directly against the perceived biological model. The results highlight a stark contrast between a strictly data-driven temporal model and a rigid parametric approach.

As visualized below, the perceived biological function for our domain of interest for RBC (red Blood Cells) is approximated to be linear and more importantly strictly increasing across the RBC domain. However, when GPR was applied to the fabricated testing set to isolate the behavior of the RBC variable, it revealed non-linear fluctuations and localized variations that the parametric model fails to capture. This could direct us to potential flaws in data handling, inherent scarcity of data, factors that we hadn’t considered like host immune responses, or allow us to make new hypotheses about the biological system.

While the perceived model assumes global uniformity, GPR captures localized variations that suggest the underlying system might be subject to feedback loops or thresholds not accounted for in the original equation. However, a crucial trade-off of GPR is its dependency on hyperparameter selection. If the signal variance σ_f^2 or length-scale l are poorly tuned, the model risks overfitting to local noise or smoothing out legitimate biological transitions.

Isolated effect of RBC on per capita growth: GPR





Perceived approximation for the limited domain of RBC

References:

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2. Murray CJ, Ikuta KS, Sharara F, Swetschinski L, Aguilar GR, Gray A, Han C, Bisignano C, Rao P, Wool E, Johnson SC. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. The lancet. 2022 Feb 12;399(10325):629-55.
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